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Attitudes and perceptions of dental students and interns toward AI in dentistry: a cross-sectional survey in a Saudi population

Abdulfatah Samih Alazmah[1*†], Hassan Hamed Kaabi[2†], AlWaleed Fahad Abushanan[1], Ahmed Sulaiman Altuwalah[3], Rafif Faisal Alshenaiber[4], Mohammed Ali Abuelqomsan[5] and Qamar Mohammadziad Hashem[5]

Abstract

Background Artificial intelligence (AI) is transforming healthcare, including dentistry, by enhancing diagnostics, treatment planning, and patient care; therefore, understanding dental students' perceptions of AI is essential for integrating AI education into dental curricula. This study aimed to assess the knowledge, attitudes, and perceptions of AI among dental students and interns in Saudi Arabia to identify gaps and provide insights that may guide future curriculum planning.

Methods Fourth- and fifth-year dental students and interns from three dental schools in Saudi Arabia completed a validated questionnaire to assess their knowledge, perceptions, and attitudes toward AI. The data were analysed using descriptive and inferential statistics, including the chi-square test with a p value < 0.05.

Results A total of 236 participants completed the survey (response rate: 86.44%) with most (95%) participants reporting familiarity with AI. Engagement in AI-related discussions varied, with higher participation among interns (85.1%) than fourth-year students (50%). AI's role in patient care was widely accepted, particularly in diagnostic imaging (70.8–76.6%) and patient referrals (54.3–61.1%). Most participants (77.8–92.9%) supported integrating AI into dental curricula but only 55.7–60.6% felt adequately prepared to work with AI tools. Ethical concerns and job displacement fears were also noted.

Conclusions Despite high interest in AI, many dental students and interns lack adequate training and confidence in its use. Structured, hands-on education and ethical guidance are needed to bridge the gap between awareness and practical readiness, ensuring responsible AI integration into dental practice.

Keywords Artificial intelligence, Dental students, Dental curriculum, Perception, Dentistry, Dental education

[†]Abdulfatah Samih Alazmah and Hassan Hamed Kaabi contributed equally to this work.

*Correspondence:
Abdulfatah Samih Alazmah
a.alazmah@psau.edu.sa
[1]Department of Pediatric Dentistry, College of Dentistry, Prince Sattam Bin Abdulaziz University, Al-Kharj 11942, Saudi Arabia
[2]Department of Oral Medicine and Diagnostic Sciences, College of Dentistry, King Saud University, P.O. Box 60169, Riyadh 11545, Saudi Arabia
[3]Department of Restorative and Prosthodontic Dental Sciences, College of Dentistry, Majmaah University, Al-Majmaah 11952, Saudi Arabia
[4]Department of Prosthetic Dental Sciences, College of Dentistry, Prince Sattam Bin Abdulaziz University, Al-Kharj 11942, Saudi Arabia
[5]Department of Conservative Dental Sciences, College of Dentistry, Prince Sattam Bin Abdulaziz University, Al-Kharj 11942, Saudi Arabia

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Background

Technology is advancing rapidly, making computer science, including artificial intelligence (AI), an essential part of daily practice. AI enables machines to analyze data and support decision-making through algorithms that mimic human reasoning and learning [1, 2]. AI is widely used in various industries, including healthcare, opening up numerous possibilities for developing high-quality services, creating advanced products, and innovating new business strategies [3].

AI is anticipated to play a significant role in enhancing human efficiency and decision-making capabilities [4] with recent advances in computer and informatics technologies enabling the integration of AI technologies, including machine learning and deep learning, into healthcare information systems [5]. AI has been widely incorporated into decision support systems (DSSs) in data-heavy medical fields such as radiology and ophthalmology [6], therefore healthcare professionals need to be educated about the use of AI to facilitate its application for improved patient care [7].

AI has been widely applied in dentistry [8, 9] and has become a valuable asset across numerous dental specialties including radiography, oral and maxillofacial surgery, orthodontics, and implantology [10, 11] for dental diagnosis [10, 12, 13], treatment planning [10, 13, 14], risk assessment, and outcome prediction [15]. Moreover, research and development in AI applications are driving substantial changes within the field. However, unlike other fields, AI technology faces challenges in replacing dentists as it cannot engage in complex discussions with patients, build trust, provide reassurance, and demonstrate empathy—key aspects of patient care [8]. A recent comprehensive study on healthcare students from different specialties, including dentistry, revealed a moderate level of knowledge toward the utilization of AI technology. However, the expertise and practical abilities are lacking, highlighting the need to incorporate AI-relevant educational programs into the academic curricula [16].

While a few studies have explored dental students' awareness and perceptions of AI in Saudi Arabia and internationally [17–19], the existing literature remains limited in scope, especially regarding students' preparedness for AI integration and its impact on clinical decision-making and education. To address these gaps, this study focuses on dental interns and undergraduate students from three major public dental schools in central Saudi Arabia, offering a targeted perspective on attitudes and perceptions in a context where AI-related curricular integration is still evolving.

Methods

Study design and setting

This cross-sectional study was conducted between February and April 2024 across three government dental colleges in central Saudi Arabia: Prince Sattam bin Abdulaziz University (PSAU), Majmaah University (MU), and Al-Qassim University (QU). These institutions were selected to represent a range of educational environments, and the study focused on academic levels most engaged in clinical training and likely to encounter AI applications in practice. The study was approved by the Institutional Review Board of Prince Sattam University (SCBR-276/2024) and conducted in accordance with the Declaration of Helsinki. Electronic informed consent was obtained from all participants at the beginning of the survey.

Study participants

Eligible participants were fourth- and fifth-year dental students and dental interns enrolled at three government dental colleges. Inclusion criteria were current enrollment in one of the participating institutions and willingness to participate with informed consent. Surveys were excluded if they were incomplete or submitted more than once. Only fully completed, unique responses were included in the final analysis, and no imputation was performed. At the time of data collection, PSAU did not enroll female students or interns, which contributed to the gender imbalance observed in the sample.

Sample size and technique

Sample size was calculated using G*Power software (version 3.1) to detect a medium effect size ($w = 0.3$) for chi-square tests across three groups (4th year, 5th year, interns), with a significance level of $\alpha = 0.05$ and power = 0.90. The minimum required sample size was 207. A total of 236 participants completed the survey, exceeding this requirement. Recruitment was facilitated via institutional email lists and official student WhatsApp groups, with the support of faculty coordinators.

Data collection

The structured questionnaire was adapted from previously validated instruments used in similar studies [17, 20, 21], with modifications to align with the Saudi dental education context. To ensure content validity, two academic experts in dental education independently reviewed the revised questionnaire for clarity, relevance, and domain coverage. A pilot test was conducted with 10 students (excluded from the final analysis) to assess comprehension and usability, leading to minor refinements. Due to the limited number of Likert-scale items per domain, internal consistency measures such as Cronbach's alpha were not calculated. All survey items were mandatory in Google Forms, ensuring that no missing responses occurred.

The final questionnaire consisted of 20 items organized into five sections: demographic information, knowledge of AI, perceptions of AI in patient care,

perceptions of AI in the health system, and perceived impact of AI on ethics and dental education. Following prior literature (e.g., Jha et al. 2022) [20], items using "likelihood" response formats (e.g., likelihood of AI assisting in patient care or hospital operations) were grouped under the "perception" domain to reflect participants' evaluative beliefs. The complete list of questionnaire items and corresponding response formats is provided in Additional file 1 (Table S1).

The response formats varied by section. Yes/No questions were used for knowledge-based items. For the remaining sections, participants responded using a five-point Likert scale. Items assessing likelihood used the options: *extremely likely*, *likely*, *uncertain*, *unlikely*, and *extremely unlikely*. Items assessing agreement used: *strongly agree*, *agree*, *neither agree nor disagree*, *disagree*, and *strongly disagree*. For data analysis, responses were combined into three categories to facilitate interpretation: *extremely likely* and *likely* were grouped together; *extremely unlikely* and *unlikely* were also grouped together; *uncertain* responses remained separate. Similarly, *strongly agree* and *agree* were combined; *disagree* and *strongly disagree* were grouped; and *neither agree nor disagree* was retained as a distinct category. The online questionnaire was distributed via Google Forms. Participation was voluntary, anonymous, and incentive-free. The estimated time to complete the survey was 8–10 min.

Statistical analysis

Survey responses were exported from Google Forms into Microsoft Excel and analyzed using IBM SPSS Statistics version 21 (Armonk, NY, USA). Descriptive statistics were used to summarize demographic and response distributions. Chi-square tests were performed to assess associations between academic level and survey responses. A p-value of < 0.05 was considered statistically significant. Given the exploratory nature of the study and the categorical structure of most items, advanced analyses such as multivariate or ordinal regression were not conducted.

Results

The survey was distributed to 273 dental students and interns across the three dental schools (Table 1), of which, 236 participants completed the survey, achieving a high response rate of 86.44%, which supports the representativeness of the responses within the sampled institutions. Most respondents were male (163; 69.1%), mainly due to the absence of female students at PSAU. All questionnaire items received complete responses, and the full distributions of Likert-scale answers are summarized in Tables 2, 3 and 4.

Individual knowledge regarding AI

There was a strong understanding of AI across all groups with < 95% indicating familiarity (Fig. 1a). Interest in machine learning was also high, particularly among interns (80.9%), followed by fifth-year (77.8%) and fourth-year students (68.6%) (Fig. 1b). Engagement with AI-related discussions decreased with educational level as follows: 85.1% of interns, 74.3% of fifth-years, and 50.0% of fourth-years reported participation (Fig. 1c). These findings indicate that while awareness of AI is high, familiarity and engagement clearly increase with academic progression, suggesting that practical experience may enhance students' understanding of AI applications.

Perception of AI toward patient care

The participant's responses about the perception of AI toward patient care in Table 2 demonstrated positive attitudes toward adopting AI across various aspects of patient care, including analyzing patient information for diagnoses, reading and interpreting diagnostic imaging, determining patient referrals, and formulating treatment plans. There were no significant differences between groups, indicating consistent perceptions across training levels. This suggests a shared openness to integrating AI into clinical workflows and recognition of its potential to enhance diagnostic accuracy and treatment planning, even among students with limited clinical experience.

Perception of AI toward the health system

The analysis highlights consistently strong positive perceptions of AI's potential in various health system applications across all training levels (Table 3). The respondents

Table 1 Response rates by academic level, gender, and dental school

	Academic level	**4th year 72 (30.5%)**		**5th year 70 (29.7%)**		**Interns 94 (39.8%)**		**Total**
	Gender*	**Female**	**Male**	**Female**	**Male**	**Female**	**Male**	
Dental school	**PSAU**	-	15	-	14	-	32	61 (25.8%)
	MU	12	20	1	16	4	15	68 (28.8%)
	QU	12	13	16	23	28	15	107 (45.3%)
	Total	24	48	17	53	32	62	**236**

*Female (73, 30.9%) vs. male (163, 69.1%). *PSAU* Prince Sattam bin Abdulaziz University, *MU* Majmaah University, *QU* Al-Qassim University

Table 2 Participants' perception of AI toward patient care

Question	4th year	5th year	Interns	
	N (%)			*p*
Use of AI to analyse patient information to reach diagnoses				
Extremely likely/likely	42 (58.4)	38 (54.3)	56 (59.6)	0.6504
Extremely unlikely/unlikely	23 (31.9)	26 (37.1)	28 (29.8)	
Uncertain	7 (9.7)	6 (8.6)	10 (10.6)	
Use of AI to read and interpret diagnostic imaging				
Extremely likely/likely	51 (70.8)	53 (75.7)	72 (76.6)	0.486
Extremely unlikely/unlikely	17 (23.6)	13 (18.6)	15 (15.9)	
Uncertain	4 (5.6)	4 (5.7)	7 (7.5)	
Use of AI to evaluate when to refer patients to specialists				
Extremely likely/likely	44 (61.1)	38 (54.3)	53 (56.4)	0.624
Extremely unlikely/unlikely	22 (30.6)	27 (38.6)	32 (34)	
Uncertain	6 (8.3)	5 (7.1)	9 (9.6)	
Use of AI to formulate a treatment plan				
Extremely likely/likely	43 (59.7)	38 (54.3)	52 (55.3)	0.712
Extremely unlikely/unlikely	23 (31.9)	27 (38.6)	30 (31.9)	
Uncertain	6 (8.4)	5 (7.1)	12 (12.8)	

Significance level < 0.05, *pp* value (chi-square test)

Table 3 Participants' perception of artificial intelligence (AI) toward the health system

Question	4th year	5th year	Interns	*p*
	N (%)			
Provide documentation (e.g., patient-updated medical records)				
Extremely likely/likely	69 (95.8)	68 (97.2)	86 (91.5)	0.129
Extremely unlikely/unlikely	2 (2.8)	1 (1.4)	7 (7.4)	
Uncertain	1 (1.4)	1 (1.4)	1 (1.1)	
Assist hospitals in capacity planning and human resource management				
Extremely likely/likely	66 (91.7)	65 (92.9)	87 (92.6)	0.936
Extremely unlikely/unlikely	4 (5.5)	3 (4.3)	5 (5.3)	
Uncertain	2 (2.8)	2 (2.8)	2 (2.1)	
Provide recommendations for quality improvement in practices/hospitals				
Extremely likely/likely	66 (91.7)	67 (95.8)	87 (92.6)	0.684
Extremely unlikely/unlikely	4 (5.5)	2 (2.8)	5 (5.3)	
Uncertain	2 (2.8)	1 (1.4)	2 (2.1)	

Total respondents ($N = 236$); 4th year: 72, 5th year: 70, Interns: 94. Significance level < 0.05, *pp* value (chi-square test)

recognized AI's value in managing and updating patient medical records, assisting in hospital capacity planning and human resource management, and enhancing quality improvement in practices and hospitals, with no significant differences between academic levels ($p = 0.684$). These findings demonstrate broad recognition of AI's role in improving efficiency and decision-making within healthcare systems, suggesting that students appreciate its institutional benefits beyond direct patient care.

Impact of AI on ethics and dental education

The responses demonstrated consistent perceptions across training levels regarding the ethical implications of AI, its influence on career decisions, and its role in dental education (Table 4). There was a moderately high perception of AI's influence on specialty selection (~65%) and AI's potential to reduce jobs (~60%), with consistent perceptions across training levels, highlighting students' concern about future employment and workforce displacement. High agreement rates were observed for AI's potential to raise ethical (~70%) and social (~69%) challenges, reflecting awareness of broader issues such as data privacy, professional accountability, and fairness in AI use. Moderate confidence was reported in the adequacy of current dental education (~58–60%), but there was strong support for including AI competencies in dental curricula and mandating AI training for all dental trainees (>75%). Together, these findings show that students not only recognize the ethical and societal implications of AI but also emphasize the need for structured education to ensure its responsible and effective integration into dental practice.

Table 4 Impact of artificial intelligence (AI) on ethics and dental education

Question	4th year	5th year	Interns	*p*
	N (%)			
AI will/already did impact my choice of specialty				
Agree/strongly agree	45 (62.5)	47 (67.2)	62 (65.9)	0.981
Disagree/strongly disagree	6 (8.3)	7 (10)	9 (9.6)	
Neither agree nor disagree	21 (29.2)	16 (22.8)	23 (24.5)	
AI will reduce the number of jobs				
Agree/strongly agree	48 (66.7)	40 (57.2)	59 (62.8)	0.157
Disagree/strongly disagree	7 (9.7)	15 (21.4)	14 (14.9)	
Neither agree nor disagree	17 (23.6)	15 (21.4)	21 (22.3)	
AI in dentistry will raise new ethical challenges				
Agree/strongly agree	48 (66.7)	56 (80)	63 (67)	0.680
Disagree/strongly disagree	2 (2.8)	1 (1.4)	3 (3.2)	
Neither agree nor disagree	22 (30.5)	13 (18.57)	28 (29.8)	
AI in dentistry will raise new social challenges				
Agree/strongly agree	49 (68.1)	52 (74.3)	61 (64.9)	0.746
Disagree/strongly disagree	4 (5.5)	2 (2.8)	5 (5.3)	
Neither agree nor disagree	19 (26.4)	16 (22.9)	28 (29.8)	
My dental education is adequately preparing me to work alongside AI tools				
Agree/strongly agree	42 (58.4)	39 (55.7)	57 (60.6)	0.516
Disagree/strongly disagree	11 (15.3)	10 (14.3)	9 (9.6)	
Neither agree nor disagree	19 (26.3)	21 (30)	28 (29.8)	
Dental training should include training on AI competencies (e.g., what is AI, how will it impact us, what are the challenges it raises)				
Agree/strongly agree	56 (77.8)	65 (92.9)	77 (81.9)	0.952
Disagree/strongly disagree	2 (2.8)	2 (2.8)	2 (2.1)	
Neither agree nor disagree	14 (19.4)	3 (4.3)	15 (16)	
Every dental trainee should be required to receive training in AI competencies				
Agree/strongly agree	54 (75)	62 (88.6)	78 (82.9)	0.410
Disagree/strongly disagree	1 (1.4)	3 (4.3)	1 (1.1)	
Neither agree nor disagree	17 (23.6)	5 (7.1)	15 (16)	

Total respondents ($N=236$); 4th year: 72, 5th year: 70, Interns: 94. Significance level <0.05, *pp* value (chi-square test)

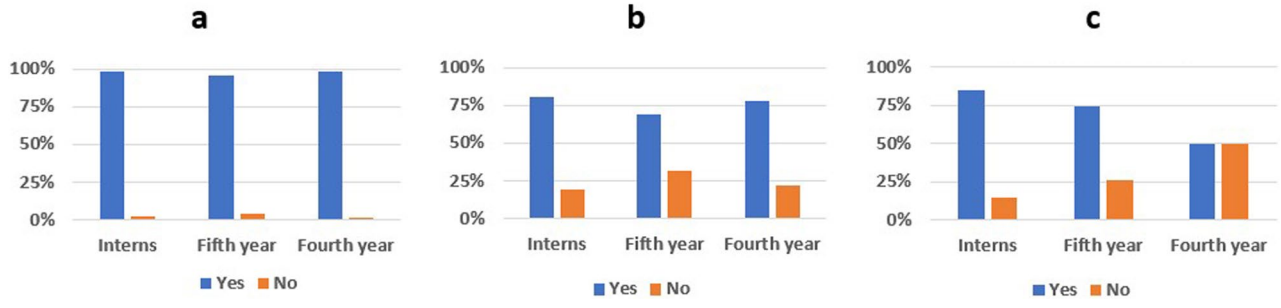

Fig. 1 Individual knowledge and engagement with artificial intelligence (AI) among groups. **a** I understand what the term "artificial intelligence" means. **b** I understand what term "machine learning" means. **c** Have you participated in or observed any presentations or discussions about AI

Discussion

AI is revolutionizing healthcare by advancing diagnostics, treatment planning, and patient care [22]. While earlier studies reported gaps in dental students' readiness for AI integration [18, 23], recent research continues to highlight the need for structured education and practical exposure [24]. This study explored perceptions of dental students and interns in Saudi Arabia and offers targeted recommendations to enhance education, preparing dental students to enhance their preparedness for AI integration.

The high response rate across three dental schools and the predominance of male participants reflect the institutional demographics of the sampled universities rather than selection bias. This aligns with prior findings of male predominance in some regions [25], although other studies have shown higher female representation [17]. This context underscores the importance of expanding AI

education opportunities across both genders and diverse institutions to ensure equitable exposure.

Overall, respondents demonstrated strong awareness of AI and positive attitudes toward its clinical applications, reflecting the growing visibility of AI in healthcare. These findings are consistent with recent Saudi and international reports [17, 19, 26, 27], suggesting that AI is no longer viewed as a futuristic concept but as an emerging component of professional practice. Unlike studies combining AI with robotics or multinational samples [23], this study provides a localized perspective, highlighting that Saudi dental students are ready to engage with AI tools. This readiness underscores an opportunity for dental schools to move from passive awareness to structured, skill-based AI education that prepares students for practical integration in clinical and diagnostic contexts.

The trend of higher AI engagement among interns suggests that clinical exposure enhances students' recognition of AI's practical relevance. Early introduction of AI-related learning experiences during preclinical years could therefore promote a smoother transition from theoretical awareness to applied competence.

Most respondents perceived AI positively in patient care, especially in diagnostics and imaging, consistent with prior findings [18, 28]. Support for its role in referral evaluation was also higher than previously reported [20], suggesting growing trust in AI as a decision-support tool rather than a replacement for professional judgment. This emerging confidence points to a broader understanding of AI's role in improving efficiency and coordination within clinical workflows.

AI's potential to assist clinicians in formulating treatment plans, particularly by predicting therapy responses, is increasingly recognized for its role in improving clinical outcomes [29]. About half of the participants endorsed this role, aligning with prior Saudi and Indian studies [17, 18, 21]. This variation across contexts likely reflects differences in students' exposure to clinical AI tools and the extent of their hands-on training.

AI is increasingly recognized for enhancing data management, resource allocation, and quality of care [30]. Respondents supported its use in administrative tasks such as record updating and capacity planning, reflecting awareness of its value in improving both clinical practice and institutional efficiency. Integrating these aspects into dental curricula could help students understand AI's broader role in modern healthcare.

Many students agreed AI would influence their specialty choice and job availability, mirroring concerns in global literature [17, 20, 23, 27, 31]. While fears about job displacement persist, evidence suggests complete replacement is unlikely due to the need for empathy, trust, and ethical care (42). Integrating ethics-based discussions and career guidance into dental curricula can help reframe AI as a supportive tool that enhances rather than replaces human expertise.

A considerable number of students recognized ethical and social challenges related to AI, such as accountability, data privacy, and transparency, reflecting alignment with global frameworks like the EU Trustworthy AI and WHO guidelines [32, 33]. This awareness may indicate readiness to engage with the ethical dimensions of emerging technologies, but also highlights the need for curricula that integrate digital ethics within professional development.

Most respondents supported integrating AI into dental education and favored mandatory training, yet only about 60% felt adequately prepared. This gap between enthusiasm and competence reflects a broader curricular shortfall also noted in previous Saudi studies [17, 18]. Despite rising awareness, limited practical exposure suggests that current programs emphasize knowledge over application. It is therefore important to integrate competency-based, hands-on AI modules, supported by case-based learning and ethical guidance, into dental curricula [34, 35]. Effective implementation of AI education requires integrating it into existing courses to ensure contextual and applied learning. Faculty development programs are also essential to equip educators with the skills and confidence to teach and assess AI-related competencies, supporting sustainable curriculum innovation in dental education [36, 37].

This study has several limitations that may affect the generalizability of its findings. The sample was drawn from three government dental schools in central Saudi Arabia and may not represent students from other regions, private institutions, or female-only colleges. The absence of these groups may have limited the diversity of educational contexts captured. Inter-institutional comparisons were also not conducted, which could have revealed differences in AI exposure or curriculum emphasis. The sample was predominantly male due to the absence of female students at PSAU, restricting gender-based interpretation. The use of self-reported data introduces potential social desirability and recall bias, as participants may have overestimated their familiarity or readiness for AI integration. Convenience sampling via email and WhatsApp may have further favored students more interested in AI or academically engaged, leading to more positive attitudes toward AI. Additionally, the study did not include stakeholder perspectives such as faculty, administrators, practicing dentists, and specialists, whose views may differ from those of students. Beyond these design constraints, the reliability of the instrument also warrants consideration. As each domain included only a few Likert-scale items, calculating Cronbach's alpha was not appropriate. Reliability was instead supported through adaptation from validated instruments, expert review, and pilot testing. Finally, some items, such

as those assessing the likelihood of using AI in patient care, may reflect behavioral intention rather than perception, introducing minor construct overlap. Future studies should adopt more representative sampling across regions, institutions, and genders, and include stakeholder perspectives to enhance generalizability. Using objective assessments alongside self-reported data and expanding item numbers with stronger reliability testing will help improve validity and provide a more robust foundation for AI curriculum development.

Conclusions

Dental students and interns expressed strong support for integrating AI into education and practice; however, only about 60% felt adequately prepared to use AI tools, revealing a clear gap between awareness and readiness. Our findings highlight the need for structured, competency-based training that combines foundational knowledge of AI and machine learning with practical application. Addressing ethical concerns and aligning curricula with evolving technologies will be key to preparing future dental professionals for responsible and effective AI adoption. Future research should define key AI competencies, assess their curricular integration through longitudinal or intervention studies, and include broader stakeholder perspectives to enhance applicability and generalizability.

Abbreviations

AI	Artificial Intelligence
IRB	Institutional Review Board
SPSS	Statistical Package for the Social Sciences
IBM	International Business Machines Corporation
DSS	Decision Support System
PSAU	Prince Sattam bin Abdulaziz University
MU	Majmaah University
QU	Al-Qassim University
WHO	World Health Organization

Supplementary Information

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Supplementary Material 1.

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Authors' contributions

AA and HK were the principal investigators and major contributors to the conceptual framework, data analysis, literature search, and manuscript writing. AT and MA collected the data. RS and QH interpreted the data. AFA prepared Fig. 1; Tables 1, 2, 3 and 4. All authors read and approved the final version of the manuscript.

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Data availability

The datasets used and/or analyzed during the current study are available from the corresponding author upon reasonable request.

Declarations

Ethics approval and consent to participate

The present retrospective study was ethically reviewed and approved by (IRB No. SCBR-276/2024; approval date: 23 April 2024). It complies with the latest version of the Declaration of Helsinki. The informed consent was also obtained from all the study participants at the beginning of the survey.

Consent for publication

Not applicable.

Competing interests

The authors declare no competing interests.

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